SPECIAL METALS

INCONEL® alloy C-276

Publication number SMC-019

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INCONEL alloy C-276 (UNS N10276/ WNr. 2.4819) is known for its corrosion resistance in a wide range of media. The high molybdenum content imparts resistance to localized corrosion such as pitting. The low carbon minimizes carbide precipitation during welding to maintain resistance to intergranular attack in heat-affected zones of welded joints. It is used in chemical processing, pollution control, pulp and paper production, industrial and municipal waste treatment and the recovery of "sour" natural gas. Applications in air pollution control include stack liners, ducts, dampers, scrubbers, stack-gas re-heaters, fans, and fan housings. In chemical processing, the alloy is used for components including heat exchangers, reaction vessels, evaporators and transfer piping.

Limiting Chemical Composition, wt %*

Ni	Balance	Mn	1.0 max.
Mo	15.0 - 17.0	C	0.01 max.
Cr	14.5 - 16.5	V	0.35 max.
Fe	4.0 - 7.0	P	0.04 max.
W	3.0 - 4.5	S	0.03 max.
Co	2.5 max.	Si	0.08 max.

*UNS N10276

Available Forms

Rod and/or Bar Products	Hot-Rolled Plate*	Cold-Rolled Sheet and/or Strip	Pipe and/or Tubing	Wire and/or Wire Rod

*Alloy-clad steel plate is also available from specialist suppliers in North America, Europe and Asia. Inquire for further details.

Specifications

Form	ASTM	ASME	ISO	Vd TUV**	DIN***
Rod and Bar	B 574	SB-574	9723	400/12.98	17752
Wire	—	—	9724	—	17753
Sheet and Plate	B 575	SB-575	6208	400/12.98	17750
Strip	B 575	SB-575	6208	400/12.98	17750
Forgings	B 564 B 366	SB-564 SB-366	9725	400/12.98	17754
Seamless Tube & Pipe	B 622 B 829*	SB-622 SB-829*	6207	400/12.98	17751
Welded Tube	B 626 B 751*	SB-626 SB-751*	—	—	—
Welded Pipe	B 619 B 775*	SB-619 SB-775*	—	—	—

*General Requirements

**VdTÜV 400/12.98 covers the temperature range -196°C (-321°F) to 450°C (842°F). At the time of publication, Special Metals is the only alloy C-276 producer with seamless tube approval.

***Chemical composition (WNr. 2.4819) is specified in DIN 17744.

Physical Properties

Density		Melting Range		Thermal Conductivity		Specific Heat	
lb/in³	0.321	°F	2415-2500	Btu•in/ft²•h•°F	67.9	Btu•lb•°F	0.102
g/cm³	8.89	°C	1325-1370	W/m•°C	9.8	J/kg•°C	427

Young's Modulus		Shear Modulus		Permeability at 200 oersted (15.9 kA/m)	Poisson's Ratio
10^6 psi	29.8	10^6 psi	11.4	1.0002	0.307
GPa	205	GPa	79		

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Physical Properties (continued)

Temperature	Coefficient of Expansion[a]	Electrical Resistivity	Young's Modulus	Temperature	Coefficient of Expansion[a]	Electrical Resistivity	Young's Modulus
°F	10^{-6} in/in•°F	ohm•cmil/ft	10^6 psi	°C	µm/m•°C	µΩ•cm	GPa
77	—	739.2	29.8	25	—	122.9	205
200	6.8	743.8	29.5	100	12.2	123.7	203
400	7.0	749.3	28.6	200	12.4	124.5	198
600	7.2	757.7	27.8	300	12.9	125.7	192
800	7.4	760.3	26.7	400	13.2	126.0	186
1000	7.5	772.5	25.7	500	13.5	127.7	180
1200	7.7	781.5	24.8	600	13.6	129.9	178
1400	8.1	773.9	23.5	700	14.1	129.7	167
1600	8.5	768.3	22.0	800	14.8	128.2	159
1800	—	766.2	20.6	900	—	127.4	150
2000	—	757.7	19.1	1000	—	127.1	141

[a] Mean coefficient of linear expansion between 77°F (25°C) and temperature shown.

Mechanical Properties

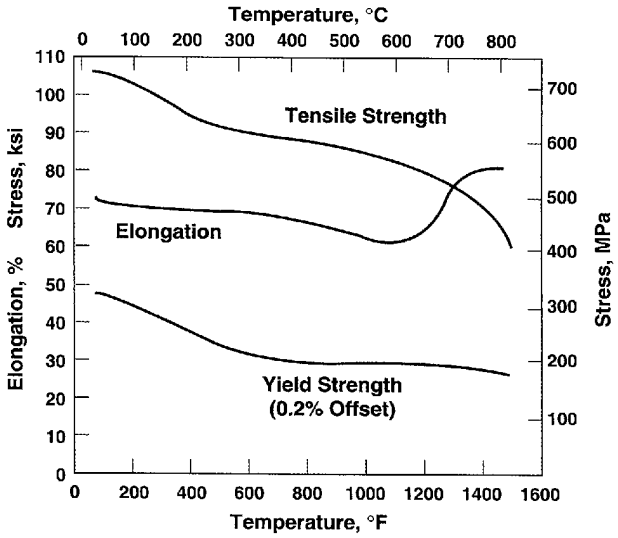
Typical Room-Temperature Tensile Properties of Annealed Material

Product Form	Tensile Strength		Yield Strength (0.2% offset)		Elongation	Hardness
	ksi	MPa	ksi	MPa	%	Rb
Tubing	105.4	727	45.4	313	70	92
Plate	107.4	741	50.3	347	67	89
Bar	110.0	758	52.6	363	62	88
Sheet	115.5	796	54.6	376	60	86

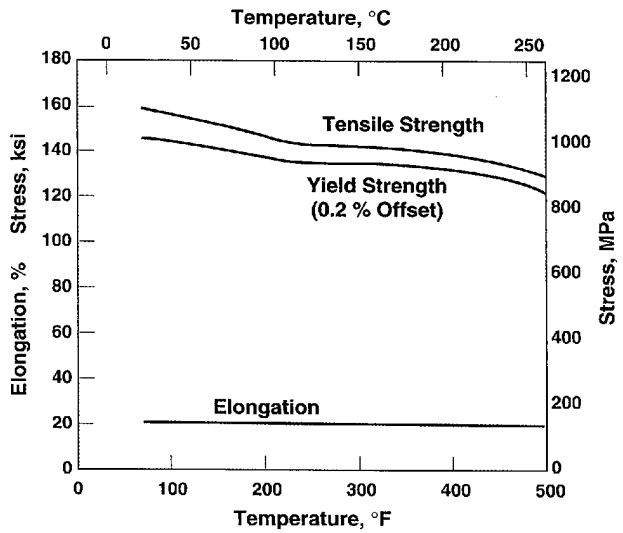
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INCONEL® alloy C-276

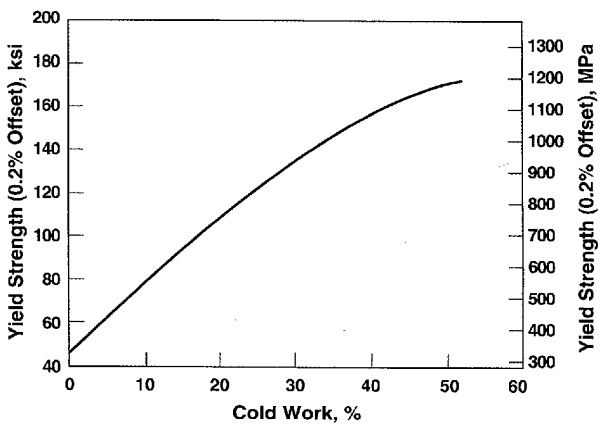
Mechanical Properties (continued)



Tensile properties of annealed plate



Tensile properties of 33.5% cold-worked tubing



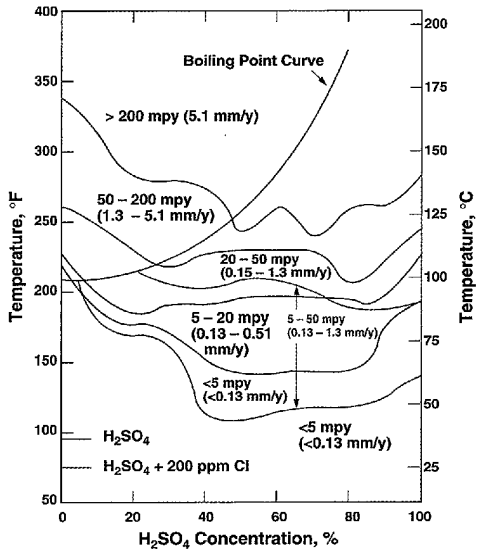
Effect of cold work on the yield strength of annealed plate

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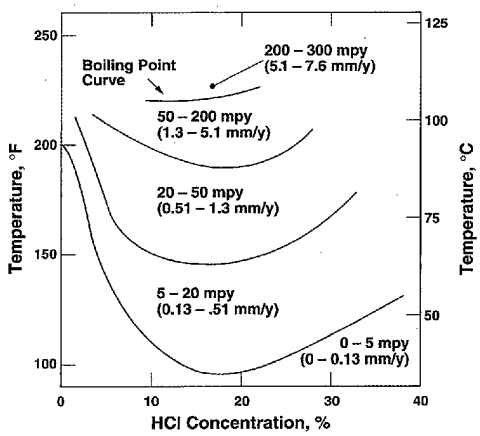
Corrosion Resistance

INCONEL alloy C-276 is resistant to general corrosion, stress-corrosion cracking, pitting and crevice corrosion in a broad range of severe environments. Its resistance to carbide precipitation during welding maintains corrosion resistance in the heat-affected zones of welded joints.

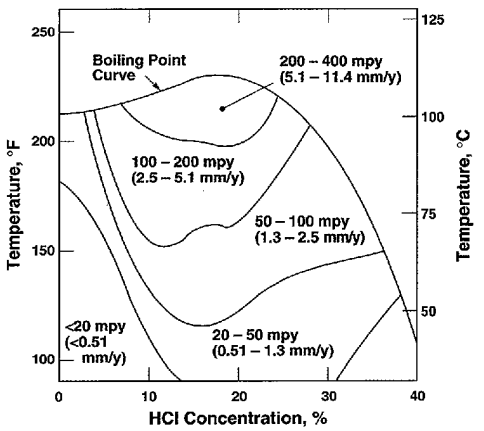
It has exceptional resistance to many of the most severe media encountered in chemical processing, including reducing and oxidizing acids, highly oxidizing neutral and acid chlorides, solvents, formic and acetic acids, acetic anhydride, wet chlorine gas, hypochlorites, and chlorine solutions. It has excellent resistance to phosphoric acid. At all temperatures below the boiling point and at concentrations lower than 65 wt%, tests have shown corrosion rates of less than 5 mpy (0.13 mm/y).



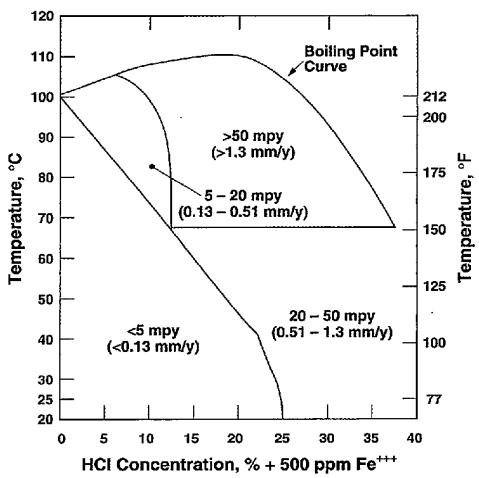
Corrosion rates in sulfuric acid



Corrosion rates in hydrochloric acid



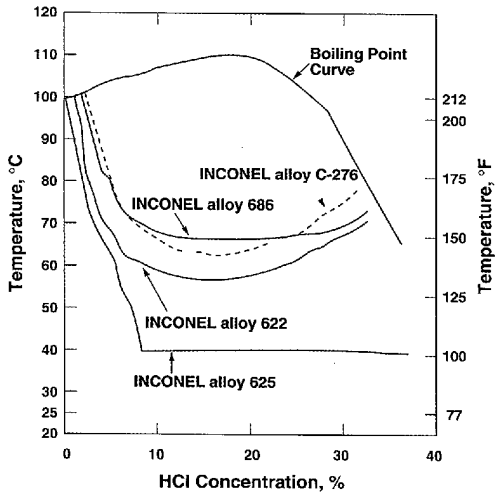
Corrosion rates in oxygen-saturated hydrochloric acid



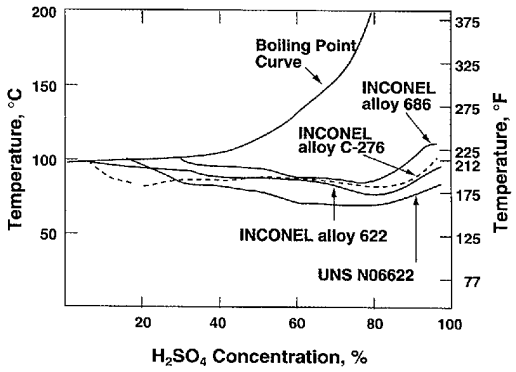
Corrosion rates in hydrochloric acid + 500 ppm Fe^{+++}

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Corrosion Resistance (continued)



Corrosion rates in hydrochloric acid



Comparative behavior of several nickel base alloys in sulfuric acid

Corrosion Rates in Acid Solutions[a]

Solution	Temperature °F	Temperature °C	Corrosion Rate, mpy (mm/y) INCONEL alloy C-276	INCONEL alloy 622[b]	INCONEL alloy 625	INCONEL alloy 686
10% H_2SO_4	Boiling	Boiling	20 (0.51)	22 (0.56)	17 (0.43)	3 (0.08)
20% H_2SO_4	176	80	3 (0.08)	1 (0.03)	1 (0.03)	—
40% H_2SO_4	176	80	5 (0.13)	10 (0.25)	5 (0.13)	—
80% H_2SO_4	176	80	4 (0.10)	9 (0.23)	6 (0.15)	4 (0.10)
5% H_2SO_4 + 0.1% HCl	Boiling	Boiling	22 (0.56)	24 (0.61)	—	—
10% H_2SO_4 + 1% HCl	Boiling	Boiling	70 (1.78)	201 (5.11)	465 (11.68)	—
10% H_2SO_4 + 2% HCl	Boiling	Boiling	138 (3.51)	281 (7.14)	—	132 (3.35)
10% H_2SO_4 + 5% HCl	Boiling	Boiling	256 (6.50)	456 (11.58)	—	—
40% H_2SO_4 + 10% HCl	176	80	26 (0.66)	32 (0.81)	—	—
2% HCl	Boiling	Boiling	43 (1.09)	52 (1.32)	—	6 (0.15)
5% HCl	140	60	10 (0.25)	—	46 (1.17)	1.2 (0.30)
20% HCl	212	100	154 (3.91)	269 (6.83)	385 (9.78)	—
5% HCl + 2% HF	158	70	18 (0.46)	40 (1.02)	102 (2.59)	—
85% H_3PO_4	Boiling	Boiling	5-25 (0.13-0.64)	23 (0.58)	>180 (>4.57)	16 (0.41)
10% HNO_3 + 3% HF	Boiling	Boiling	95 (2.41)	23 (0.61)	28 (0.71)	—

[a] 168-h tests. [b] INCONEL alloy 622 is not for sale in all countries. Inquire for availability.

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Corrosion Resistance (continued)

Corrosion Rates in Hydrochloric, Phosphoric and Acetic Acids[a]

Solution	Temperature		Corrosion Rate, mpy (mm/y)			
	°F	°C	INCONEL alloy C-276	INCOLOY alloy 25-6MO	INCONEL alloy 622	INCONEL alloy 686
0.2% HCl	Boiling	Boiling	0.60 (0.02)	<0.1 (<0.003)	<0.1 (<0.003)	0.20 (0.005)
1% HCl	Boiling 194 158	Boiling 90 70	13.3 (0.34) 6.6 (0.17) 0.74 (0.02)	119 (3.02) 37.0 (0.94) 0.02 (<0.001)	2.7 (0.07) — —	2.0 (0.05) — —
5% HCl	158 122	70 50	13.2 (0.34) 3.7 (0.09)	142 (3.61) 43.4 (1.10)	18.8 (0.48) 4.7 (0.12)	9.8 (0.25) 1.7 (0.04)
85% H_3PO_4	Boiling 194	Boiling 90	10.4 (0.26) 0.20 (0.005)	114 (2.90) 10.6 (0.27)	13.0 (0.33) 0.21 (0.005)	16.2 (0.41) 0.18 (0.005)
80% CH_3COOH	Boiling	Boiling	0.15 (0.004)	<0.1 (<0.003)	<0.1 (<0.003)	<0.1 (<0.003)

[a] 192-h tests.

Corrosion Rates in Various Media[a]

Solution	Temperature		Corrosion Rate	
	°F	°C	mpy	mm/y
10% HNO_3	Boiling	Boiling	15[b]	0.38[b]
10% HNO_3 + 3% HF	140	60	113	2.87
15% HNO_3 + 3% HF	140	60	179	4.55
20% HNO_3 + 2% HF	140	60	215	5.46
3% HF	176	80	53	1.35
10% HF	75	24	2	0.05
10% HF	176	80	28	0.71
Concentrated HF	75	24	1	0.03
Concentrated HF	176	80	34	0.86
20% H_3PO_4	Boiling	Boiling	<1	<0.03
60% H_3PO_4	Boiling	Boiling	1	0.03
85% H_3PO_4	212	100	5	0.13
85% H_3PO_4	Boiling	Boiling	121	3.07
99.9% CH_3COOH + 0.1% NaCl	Boiling	Boiling	<1	<0.03
50% NaOH	Boiling	Boiling	1	0.03

[a] Test duration of 168 h except as noted.
[b] Test duration of 24 h.

ASTM G28 Tests For Intergranular Attack

Alloy	Method A[a] Corrosion Rate		Method B[b] Corrosion Rate	
	mpy	mm/y	mpy	mm/y
INCONEL alloy C-276	210	5.33	40	1.02

[a] Boiling ferric sulfate/50% sulfuric acid.
[b] Boiling 23% H_2SO_4 + 1.2% HCl + 1% $FeCl_3$ + 1% $CuCl_2$.

The performance of INCONEL alloy C-276 in a severe test for susceptibility to intergranular attack (ASTM G28) is shown above. The base corrosion rates listed are representative of typical production material. Rates significantly higher than these indicate susceptibility to intergranular attack. This test is designed to verify mill production only and not to compare alloys for use in applications such as flue gas desulfurization.

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Corrosion Resistance – Flue Gas Desulfurization

INCONEL alloy C-276 is useful for flue gas desulfurization (FGD) systems to control air pollution from electric power plants. The alloy is used for various applications including scrubbers, ducting and stack liners.

Scrubber liquors and gas condensates generally contain chlorides, and the chloride level often determines the corrosion behavior of the materials. INCONEL alloy C-276 has been shown to withstand higher chloride content than other alloys before the onset of localized corrosion in a simulated scrubber environment.

Corrosion Rates[a] in Simulated FGD Mixed-Gas Condensate Solutions

Solution	Temperature		Corrosion Rate, mpy (mm/y)		
	°F	°C	INCONEL alloy C-276	INCONEL alloy 622	INCONEL alloy 625
Solution 1[b]	185	85	82 (2.08)	20 (0.51)	14 (0.36)
Solution 2[c]	176	80	42 (1.07)	50 (1.27)	126 (3.20)

[a] 168-h test.
[b] 60% H_2SO_4 + 0.5% HCl + 0.1% HF + 0.1% HNO_3.
[c] 60% H_2SO_4 + 2.5% HCl + 0.2% HF + 0.5% fly ash.

Maximum Pitting or Crevice Attack, mils (mm), in FGD Scrubber Slurry[a]

Alloy	Quencher	Absorber	Absorber Outlet	Outlet Duct	Bypass Duct
AISI 316L	22 (0.56)	21 (0.53)	35 (0.89)[b]	35 (0.89)[b]	12 (0.30)
AISI 317LM	20 (0.51)	22 (0.56)	29 (0.74)	33 (0.84)	29 (0.74)
INCOLOY alloy 825	15 (0.38)	33 (0.84)	39 (0.99)	50 (1.27)[b]	10 (0.25)
INCONEL alloy 625	<2 (<0.05)	10 (0.25)	11 (0.28)	7 (0.18)	nil
INCONEL alloy C-276	nil	nil	<2 (<0.05)	nil	nil

[a] 6-month exposure at 126°F (52°C), pH 5.5, 5000 ppm chlorides.
[b] Perforated.

Maximum Pitting or Crevice Attack, mils (mm), in Scrubber Slurry[a]

Alloy	Scrubber Bottom	Under Spray Nozzles	Scrubber Outlet	Hold Tank
AISI 316	5 (0.13)	7 (0.18)	49 (1.24)[b]	2 (0.05)
INCOLOY alloy 825	<2 (<0.05)	1.2 (0.03)	49 (1.24)[b]	nil
INCONEL alloy 625	nil	nil	26 (0.66)	nil
INCONEL alloy C-276	nil	nil	nil	nil

[a] 3-month exposure at 120°F (49°C), pH 5.8-6.1, 10,000 ppm chlorides.
[b] Perforated.

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Corrosion Resistance – Oilfield Applications

INCONEL alloy C-276 is one of the premier materials for recovery and handling of "sour" natural gas, which contains hydrogen sulfide and usually carbon dioxide and chlorides. The gas can be extremely corrosive to carbon and alloy steels, and may cause brittle failure of many alloys by sulfide stress cracking (hydrogen embrittlement) or stress-corrosion cracking. The high levels of nickel, chromium, and molybdenum in INCONEL alloy C-276 make the alloy resistant to sour environments even at high temperatures in deep wells. The alloy is used for tubing and a variety of other downhole and surface components.

C-Ring Tests in NACE Solution[a]

Material Condition	Simulated Well Age	Yield Strength (0.2% offset)		Hardness, Rockwell C	Duration, Days	Sulfide Stress Cracking
		ksi	MPa			
Cold Worked	600°F (315°C)/1000 h	126.6	873	32	43	No
Cold Worked	600°F (315°C)/1000 h	155.1	1069	38	43	No
Cold Worked	600°F (315°C)/1000 h	166.8	1150	35	43	No
Cold Worked	600°F (315°C)/1000 h	188.7	1301	43	43	No

[a] Room-temperature tests at 100% of yield strength in 5% NaCl plus 0.5% acetic acid saturated with H_2S. All specimens were coupled to carbon steel.

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INCONEL® alloy C-276

Welding

INCONEL alloy C-276 has good weldability and can be used as-welded for most applications. INCO-WELD filler metal and welding electrode 686CPT can be used to "overmatch" INCONEL alloy C-276 where enhanced corrosion resistance is required.

Recommended Welding Products

Shielded Metal Arc Welding	Gas Tungsten Arc Welding, Gas Metal Arc Welding
INCONEL welding electrode C-276	INCONEL filler metal C-276
INCO-WELD welding electrode 686CPT	INCO-WELD filler metal 686CPT

Environment	Base Metal Alloy (5-6 mm thickness)	Weld Filler Metal	Maximum Pitting Depth of Attack, mm Average Results for Duplicate Specimens			
			GTAW Process		GMAW Pulsed Process	
			Base Metal	Weld Metal	Base Metal	Weld Metal
Green Death*	C-276	C-276	0	6.2	0	3.4
	C-276	686CPT	0	0	0	0
ASTM G48C	C-276	C-276	0	0	0	0
	C-276	686CPT	0	0	0	0

*11.9% H_2SO_4 + 1.3% HCl + 1% $FeCl_3$ + 1% $CuCl_2$ boiling at 103°C for 72 h.

Heat Treatments

Hot forming should be between 1600 and 2250°F (870 and 1230°C), with all heavy forming above 2000°F (1090°C). INCONEL alloy C-276 is normally annealed at 2100-2150°F (1150-1175°C). Cooling should be by water quench.

Machining

Recommended Conditions for Turning with Single-Point Tools

Hardness	Depth of Cut		High Speed Steel				Tool	Carbide						Tool
			Surface Speed		Feed			Surface Speed				Feed		
								Brazed Tool		Throw Away				
	in	mm	fpm	m/min	ipr	mm/rev		fpm	m/min	fpm	m/min	ipr	m/rev	
85 R_b	0.25	6.35	12-18	3.7-5.5	1/100	1/4	T-5	30-40	10-12	40-60	12-18	1/100	1/4	C-2
45 R_c	0.05	1.27	15-20	4.6-6.1	2/250	1/5	M-36	40-50	12-15	50-100	15-30	2/250	1/5	C-2

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SPECIAL METALS

INCONEL® alloy HX

INCONEL alloy HX (UNS N06002/W.Nr.2.4665) is a high-temperature, matrix-stiffened, nickel-chromium-iron-molybdenum alloy with outstanding oxidation-resistance, and exceptional strength at up to 2200°F (1200°C).

It is used for components such as combustion chambers, afterburners and tail pipes in aircraft and land-based gas turbine engines; for fans, roller hearths and support members in industrial furnaces, and in nuclear engineering.

INCONEL alloy HX is readily fabricated and welded.

Table 1 – Nominal Chemical Composition, % by Weight

Carbon	0.1
Chromium	21.8
Cobalt	1.5
Iron	18.5
Manganese	0.5
Molybdenum	9.0
Nickel	Balance*
Silicon	0.5
Tungsten	0.6

*Reference to the balance of the alloy's composition does not guarantee this is exclusively of the element mentioned but that it predominates and others are present only in minimal quantities.

Physical Properties

Some physical properties for INCONEL alloy HX are given in Table 2. Thermal conductivity data (Table 3) and electrical resistivity (Table 5) were determined from 5 casts of cold-rolled sheet, 0.039-0.067 in (1-1.7 mm) thick, heat-treated at 2150°F (1175°C)/rapid air cooled.

Mean coefficient of linear thermal expansion data (Table 4) are for solution-treated material. A variation of ±5% can be expected for compositional changes within the release specification, processing history and form.

The dynamic moduli data (Table 6) were obtained from cold-rolled sheet, 0.039-0.067 in (1-1.7 mm) thick, heat-treated at 2150°F (1175°C)/rapid air cooled and vibrated in the flexural mode.

Table 2 – Physical Properties

Density, g/cm³	8.22
lb/in³	0.297
Melting Range, °C	1260-1355
°F	2300-2470
Specific Heat, J/kg °C	461 at 20°C
Btu/lb °F	0.110 at 70°F
Poisson's Ratio at room temperature	0.320

Available Forms

INCONEL alloy HX is available as rod, bar, billet, extruded section, plate, sheet, strip and wire.

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Data contained in this publication are typical of the properties listed but are not suitable for specifications unless given as limiting.

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Table 3 – Thermal Conductivity

Temperature		Thermal Conductivity	
°C	°F	W/m °C	Btu/ft h °F
20	68	11.6	6.68
100	212	12.9	7.48
200	392	14.6	8.46
300	572	16.3	9.44
400	752	17.9	10.4
500	932	19.5	11.3
600	1112	21.1	12.2
700	1292	22.9	13.2
800	1472	24.6	14.2
900	1652	26.3	15.2
1000	1832	27.9	16.1

Table 4 – Mean Coefficient of Linear Thermal Expansion

°C	10^{-6}/°C	°F	10^{-6}/°F
20 – 100	13.3	70 – 200	7.40
– 200	14.0	– 600	7.96
– 300	14.3	–1000	8.27
– 400	14.5	–1200	8.53
– 500	14.7	–1350	8.69
– 600	15.1	–1500	8.88
– 700	15.7	–1600	9.01
– 800	16.0		
– 900	16.3		
–1000	16.7		

Table 5 – Electrical Resistivity

Temperature		Electrical Resistivity	
°C	°F	μΩ cm	Ω circ mil/ft
20	68	116	698
100	212	118	710
200	392	120	722
300	572	122	734
400	752	124	746
500	932	126	758
600	1112	127	764
700	1292	127	764
800	1472	127	764
900	1652	127	764
1000	1832	128	770

Table 6 – Dynamic Moduli

Temperature		Dynamic Young's Modulus		Dynamic Torsional Modulus	
°C	°F	GPa	10^6 psi	GPa	10^6 psi
20	68	205	29.7	80	11.6
100	212	202	29.3	78	11.3
200	392	195	28.3	75	10.9
300	572	190	27.6	72	10.4
400	752	183	26.5	70	10.2
500	932	177	25.7	66	9.6
600	1112	168	24.4	62	9.0
700	1292	161	23.3	60	8.7
800	1472	153	22.2	56	8.1
900	1652	145	21.0	53	7.7
1000	1832	135	19.6	49	7.1

Mechanical Properties

Room-temperature mechanical properties, typical of a single representative heat are quoted in Table 7. Data in Fig. 1 are for cold-rolled sheet, 0.039-0.067 in (1-1.7 mm) thick, heat-treated at 2150°F (1175°C)/rapid air cooled. Strain rates of 0.005/min to proof stress (at room temperature), 0.002/min to proof stress (at elevated temperatures), and 0.1 min thereafter.

Data in Fig. 2 are for cold-rolled, welded sheet, 0.039-0.067 in (1-1.7 mm) thick, heat-treated at 2150°F (1175°C)/rapid air cooled. The sheet was welded by the gas-tungsten-arc (T.I.G.) process using INCONEL filler metal HX with argon as the shielding gas.

Creep-rupture properties of sheet (heat-treated at 2150°F (1175°C)/rapid air cooled) are shown in Fig. 3 by Larson-Miller presentation.

Table 7 – Mechanical Properties of Sheet at Room Temperature

Tensile Strength		0.2% Proof Stress		Elongation	Hardness
ksi	MPa	ksi	MPa	%	Rb
115	793	50	345	45.5	90

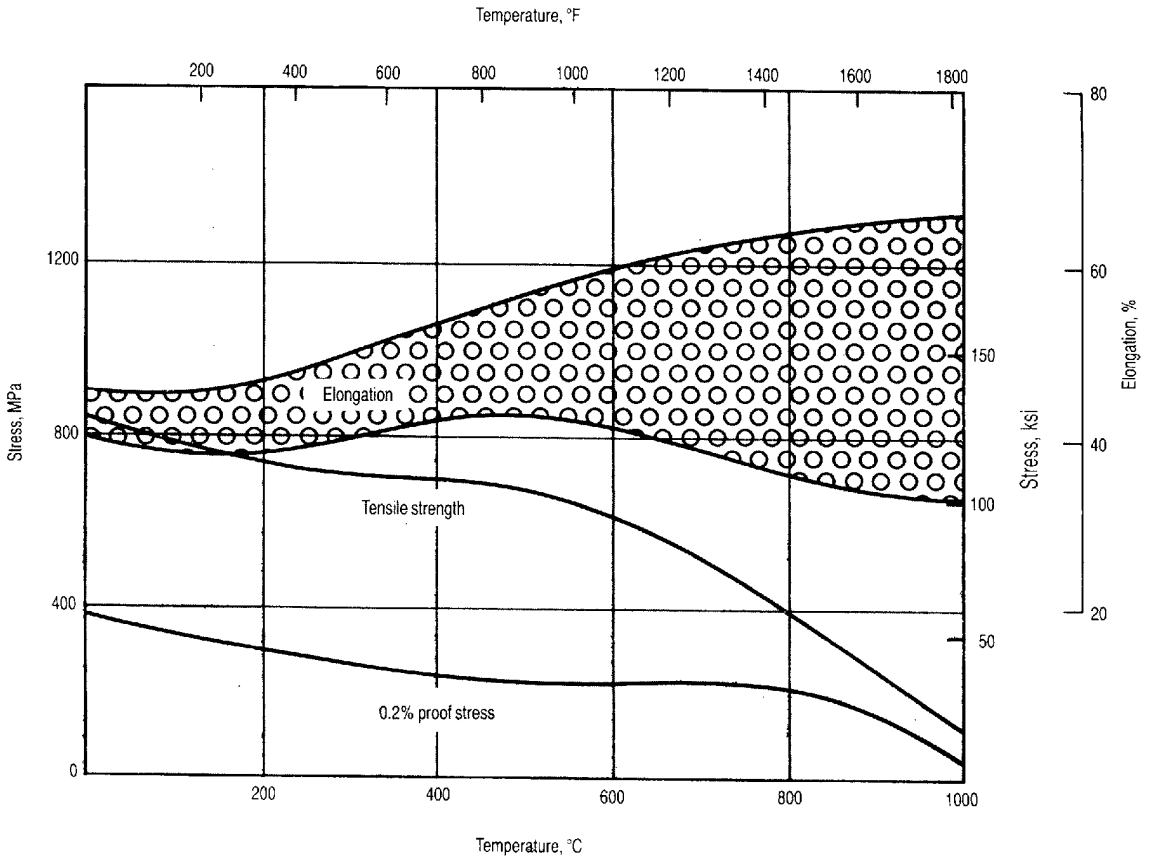


Figure 1. Tensile properties of INCONEL alloy HX sheet.

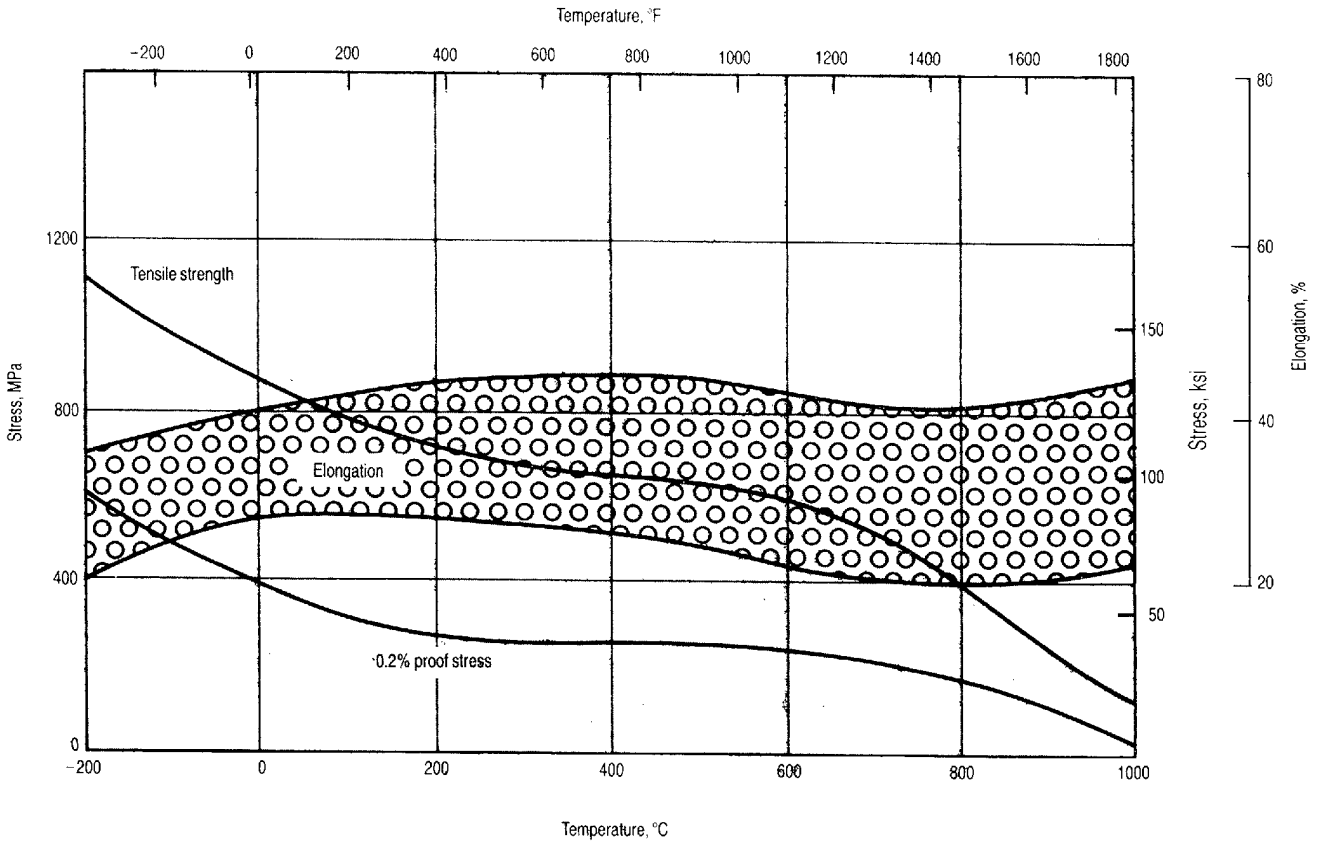
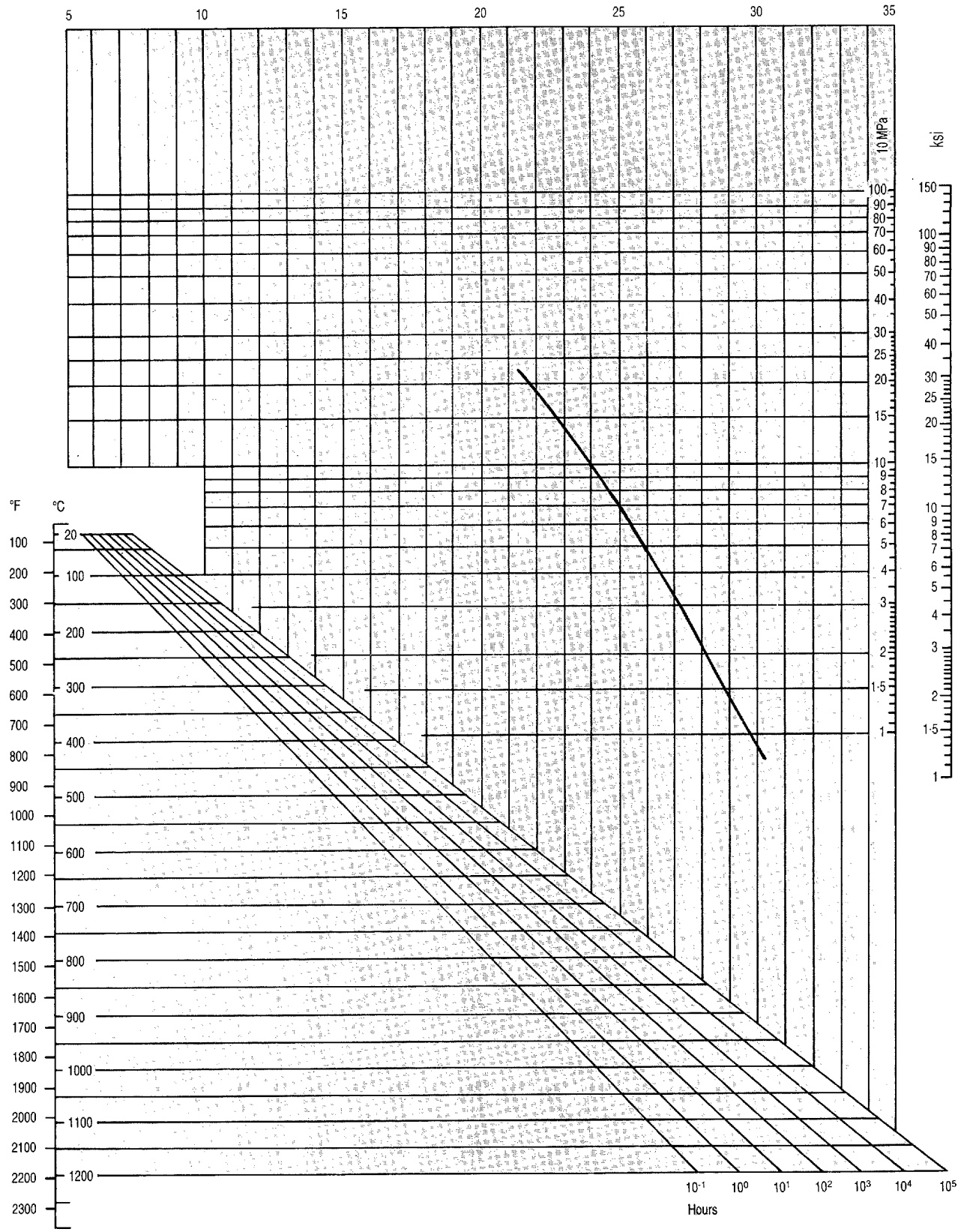


Figure 2. Tensile properties of INCONEL alloy HX welded sheet.

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Figure 3. Creep-rupture properties of INCONEL alloy HX sheet.

Larson-Miller Parameter, $T(20 + \log t) \times 10^{-3}$; T in °K, t in hours



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Corrosion-Resistance

INCONEL alloy HX has outstanding oxidation-resistance at temperatures up to 2200°F (1200°C). Test results at 1920 and 2010°F (1050 and 1100°C) for 100 hours' continuous exposure are shown in Fig. 4.

In a carburizing atmosphere of 2% methane plus 98% hydrogen flowing at 3 furnace volumes per hour for 100 hours at 1800°F (980°C), samples of a representative heat of INCONEL alloy HX showed an average weight gain of 7.7 mg/cm².

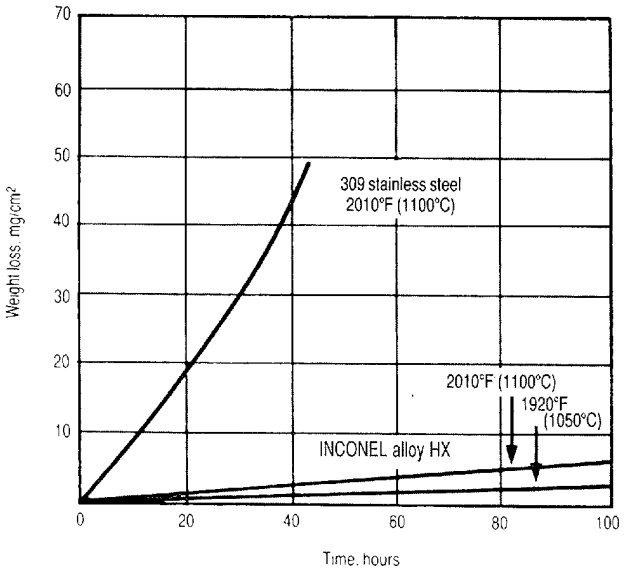


Figure 4. Comparative oxidation-resistance of INCONEL alloy HX and AISI 309 stainless steel

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Fabrication

HEAT TREATMENT

The recommended heat treatment is 2150°F (1175°C), followed by rapid air cooling or water quenching. INCONEL alloy HX is normally supplied in this solution-heat-treated condition. For some applications it can be given other treatments. Consult Special Metals for further details.

HOT AND COLD FORMING

INCONEL alloy HX can be readily cold formed using slightly higher power than that used for austenitic stainless steels. It is best cold formed in the solution-heat-treated condition.

For hot forming, the alloy should be heated to a starting temperature of 2150°F (1175°C). It should not be hot worked below 1800°F (980°C).

WELDING

The alloy can be welded by metal arc, gas-tungsten-arc (T.I.G.) and gas-metal-arc (M.I.G.) and submerged arc processes. Welding surfaces must be thoroughly descaled and cleaned. Welding is best carried out flat since the fluidity of the alloy makes position welding difficult.

INCONEL welding electrode 117 and INCONEL filler metal 617 produce weldments that match or exceed base metal high-temperature strength and oxidation-resistance. Matched composition welding consumables (INCONEL filler metal HX and INCONEL welding electrode HX) are available.

Pre-heating and post-weld heat treatments are not normally necessary. For the greatest corrosion-resistance, however, solution-heat-treatment after welding may be necessary.

MACHINING

INCONEL alloy HX should be machined in the annealed condition using carbide or high-speed steel tools and a copious supply of sulfur-free cutting fluid. Tools and machines should be as rigid as possible to minimize deflection and vibration. Speeds and feeds will be similar to those for NIMONIC alloy 75.

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Specifications

The following are applicable to INCONEL alloy HX:–

ASTM	
B 435	Plate, sheet and strip
B 572	Rod
B 619	Welded pipe
B 622	Seamless pipe and tube
B 626	Welded tube
SAE	
AMS 5536	Sheet, strip and plate
AMS 5587	Seamless tube
AMS 5588	Welded tube
AMS 5754	Bar, forgings and rings
BSI	
HR6	Billet, bar and forgings
HR204	Plate, sheet and strip
Deutschen Luftfahrt	
LW2.4665 Part 1	Sheet and strip
LW2.4665 Part 2	Bar and forgings
Ministère d'Etat de la Défense	
AIR 9165-43	Bar, forgings and sheet
AECMA	
PrEN2182	Bar
PrEN2182	Forgings
PrEN2184	Section for welded rings
PrEN2185	Sheet, strip and plate
DIN	NiCr22Fe18Mo

Sheet is also available to the following commercial specifications:–

General Electric	B50A436-S5 and B50TF24-S4
Pratt & Whitney Aircraft	1038-G
Westinghouse	PDS 15102QEC
SNECMA	DMD491.23
SAAB	AS 11736

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